
VIDEO MANAGEMENT SYSTEM INITIATIVE

PRODUCT REQUIREMENTS & USE CASE DOCUMENT

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1 INTRODUCTION

1.1 EXECUTIVE SUMMARY

The Platform Company (TPC) currently provides video management and playback/ usage services for Disney Studio All Access (DSAA). The business has decided to replace The Platform Company and retain the functionality that TPC currently provides via other channels. This document explains and evaluates the possible options (i.e. CSG as a turnkey solution, Shared Services Solution, Phased Change Approach) for a replacement VMS and player to improve upon the foundation of DSAA. In conclusion, a recommendation of the best option is made based on discussions and information available till date.

1.2 OPTIONS OVERVIEW

The most viable options for replacing TPC that are listed below:

1. **CSG as a turnkey solution** – CSG would provide both video management and playback services and would essentially replace TPC in entirety
2. **Shared Services Solution (RECOMMENDED)** – the features to be replaced would be divided between CSG and Disney and hence video management would be handled in-house whereas playback/ usage services would be provided by CSG. The entire change is planned to happen before DSAA Go-Live.
3. **Phased Change Approach** – the move away from TPC would be broken down into phases:
 - o **Phase 1:** VMS is moved in-house and TPC continues to provide playback/ usage services
 - o **Phase 2:** playback/ usage services are moved out of TPC and to another vendor (CSG)
4. **In-house Hosted Solution** – In addition to moving the VMS in-house, playback/ usage services would be all hosted in-house and there would be no third-party dependencies

1.3 OBJECTIVES

The business objectives for this initiative are:

- Increase control and ownership of VMS functionality
- Create a highly scalable VMS that is more tailored to DSAA's current and future business requirements
- Reduce future dependence on an unresponsive vendor
- Strengthen the integration of the VMS to all the other systems supporting DSAA – KeyChest, CMS, STORM, e-commerce, etc.
- Improve production workflow and efficiency

Why are we replacing The Platform?

Some of the pain-points associated with the Platform Company are as follows:

- **Inefficient Publishing Workflow** : Updating any file in a media object requires un-publishing and re-publishing of the entire media object
- **Low visibility:** Since the system is not hosted in-house and because of TPC practices, there is almost no visibility to the data or code in TPC
- **Sync issues with entitlements** : TPC stores a local copy of the entitlements that is supposed to mirror Keychest, but issues with the local copy cause recurring issues
- **Delayed feature changes/additions** : Feature changes/additions and bug fixes are not being completed in a timely manner
- **Feeds, rights, & usage rules do not propagate through environments** and have to be recreated in every environment every time there is an addition or a change

What is the timeline?

The timeline for this initiative will depend on the solution that is chosen. Ideally, the solution is implemented before DSAA launch. The choice of solution also depends on impacts of TPC replacement on other systems and how the impacts will be handled.

Which systems will be affected?

On a very high level, the CMS, SOLR, Keychest, DSAA, and all API layers will be affected at some level or another by TPC replacement. (Details in section 2.2)

- Minimize VMS errors and bugs that affect production
- Allow for development of new VMS functionality to be easily accessible in-house

1.4 FEATURE MATRIX

Given below is a list of all the features along with short descriptions and columns categorizing and prioritizing each of the features. The TPC column shows the features available for use today and the CSG column shows whether a feature is available with them or not. Medusa 2.0 is an in-house video asset creation and DRM packaging system under development currently (and planned to be complete by July). The column shows if a feature would be available when Medusa is complete and in production.

#	Features	Description	Category	Priority	TPC	CSG	Medusa 2.0
1	Media object creation - Mezzanine file ingestion	Initial creation of an "object" for a title that includes the location of the mezzanine file and all other components related to the product	Video Mgmt	P1	Y	N	Y
2	Technical metadata ingestion from STORM	Ability to import technical title metadata (chapter marks, subtitles) from Disney's internal metadata repository	Video Mgmt	P1	Y	N	Y
3	Publishing videos to transcode jobs	Authorized users can transcode mezzanine source files into multiple compressed formats with varying bit-rates	Video Mgmt	P1	Y	N	Y
4	Video queue management	MPX provides a single work progress queue for any publishing operation: transcodes, image uploads, DRM packaging...etc. all in a single window. Behind the scenes, MPX is able to load balance between the transcode nodes.	Video Mgmt	P1	Y	N	y
5	DRM packaging process - Flash Access	ThePlatform has provided plugins to RMP for various packaging servers. These plugins allow RMP to communicate with each DRM packaging server to trigger packaging jobs, extract the necessary info (different for each DRM), then upload them back into MPX once the packaging is done	Video Mgmt	P1	Y	N	Y
6	DRM packaging process - PlayReady	ThePlatform has provided plugins to RMP for various packaging servers. These plugins allow RMP to communicate with each DRM packaging server to trigger packaging jobs, extract the necessary info (different for each DRM), then upload them back into MPX once the packaging is done	Video Mgmt	P1	Y	N	Y
7	DRM packaging process - Widevine	ThePlatform has provided plugins to RMP for various packaging servers. These plugins allow RMP to communicate with each DRM packaging server to trigger packaging jobs, extract the necessary info (different for each DRM), then upload them back into MPX once the packaging is done	Video Mgmt	P1	Y	N	Y
8	Append DRM package to media object	Whenever RMP finishes a publishing job, the outcome of the operation will always be appended to the relevant media object. This applies to transcode jobs, packaging jobs, and other tasks.	Video Mgmt	P1	Y	N	Y
9	Upload encrypted assets	Upload the DRM packages to external CDNs. Batch create public URLs to these encrypted assets to each media object in MPX.	Video Mgmt	P1	Y (poor)	N	Y
10	Populate VMS with encrypted asset	Gather and upload data per encrypted asset to the VMS.	Video Mgmt	P1	Y	N	Y
11	Associate static image assets	Producers can relate static image assets (ie. thumbnails, poster art) to a media object	Metadata	P1	Y	Y	n/a

#	Features	Description	Category	Priority	TPC	CSG	Medusa 2.0
12	Inter-dependency/connection of media objects	Producers can associate bonus materials and trailers to a feature	Metadata	P1	Y	Y	n/a
13	Publish workflow	Producers are able to promote media object of record to other environments and able to publish media objects and metadata across all environments	Publishing	P1	Y	Y	n/a
14	Feeds and Categories	Producers can create and manage feeds and categories for all media objects. Ability to organize Feature titles for the consumer under categories provided by marketing.	Metadata	P1	Y	Y	n/a
15	Web based VMS	Users can access the VMS via a web browser.	General	P1	Y	Y	n/a
16	Entitlement management	The VMS will have the ability to directly connect to KeyChest to obtain user entitlements.	Usage Mgmt	P1	Y (local copy)	Y (direct)	n/a
17	License servers	A license server will be available to validate video licenses for playback.	Usage Mgmt	P1	Y	Y	n/a
18	Business rules / policy management	Business rules and policies related to usage can be set and enforced.	Usage Mgmt	P1	Y	?	n/a
19	Concurrency policy	Rules related to concurrent usage can be set and enforced. (Disney requires user to be able to stream movies using a single account on 5 different devices and be able to stream one movie on three devices at the same time)	Usage Mgmt	P1	Y	Y (7/25)	n/a
20	Geo-filtering	Usage rules specific to geographical locations can be set and enforced.	Usage Mgmt	P1	Y	Y	n/a
21	Internationalization support	Country and region specific usage rules (including support for language and currencies) can be set and enforced.	Usage Mgmt	P2	?	Y	n/a
22	Video playback - Chaptering	The user will be able to see chapter keypoints in the playback control bar of the video player and jump to any chapter in the title.	Playback	P2	Y	Y	n/a
23	Video playback - Closed Captioning	The user can enable closed captioning for the video they watch	Playback	P1	Y	Y	n/a
24	Video playback - Subtitles	The user can enable and disable subtitles while watching the video	Playback	P1	Y	Y	n/a
25	Video playback - Bookmarking/Resume	The user will be able to interact with the video player to resume the video at a point where they previously stopped.	Playback	P1	Y	Y	n/a
26	Video playback - Basic functions	The user will be able to interact with the video player to perform basic operations of play, pause, rewind, fast forward, scrubbing, full screen.	Playback	P1	Y	Y	n/a
Nice to Have Features							
27	Analytics for video playback	The tracking of video player analytics would be supported.	Analytics	P1	Y	Y	n/a
28	Batch upload of image assets	Producers would be able to work efficiently by batch uploading media assets.	Metadata	P1	Y	TBD	n/a
29	Media asset sorting	Producers would be able to manage and sort metadata and media assets with greater ease and efficiency.	Metadata	P1	N	Y	n/a
30	Rich Media file types	The business would be able to serve up and support rich media file types beyond just video (ie. Games).	Metadata	P2	Y	Y	n/a
31	Livestreams	The player should be able to support live streaming of media	Playback	P2	?	?	n/a

1.5 TPC IMPACTS/ TOUCHPOINTS

Given below is a list of components that TPC touches today and the level of impact it has on the component. The components are part of three layers as shown in the table below – Client, API, and Data. The Change required column shows if TPC replacement will require this component to be replaced, updated, or not affect the component at all.

Layer	Component	Impact Level	Change Required	Notes
Client	Web Player	High	Replace	Flash Player for Web needs to be replaced
Client	Concurrency Locking	Medium	Update	Does not impact Web but other clients need to update how a lock is acquired and associated heartbeat
Client	Playback Profile	Low	Update	Web needs ability to retrieve bookmarks but this may be easily put behind the API
Client	Playback License	Low	Update	Xbox retrieves license directly but this may be handled by using updated packaged files
API	Video Catalog	High	Replace	Requires extensive rewrite of entire API
API	Marketing	High	Replace	Will require rewrite of entire API if we merge CMS with VMS
API	Account Management	Low	None	No change should be required but due to dependencies there may be minimal impact
API	Guest Profile	Low	None	No change should be required but due to dependencies there may be minimal impact
Data	Metadata Import/Export	Medium	Replace	Various scripts are used to manually move data between systems
Data	Entitlement Sync	Medium	Update	Sync of entitlements from Keychest is initiated by TPC but this may not be required if we use Keychest directly
Data	Search Data Sync	High	Replace	Nightly data dump of all metadata to be imported into SOLR for search

2. OPTIONS ANALYSIS

With the background of feature requirements and current impacts of TPC, given below is a detailed description of each of the options mentioned in the introduction section along with integration requirements and risks/gaps that will be introduced should we decide to go with the option.

2.1. CSG AS A TURNKEY SOLUTION

This solution plans that CSG would provide **both video management and playback services** and would essentially replace TPC in its entirety. This option would eliminate the need for Disney having to build something in-house but at the same time it introduces the risk of CSG having to build features to make up for the gaps that are shown in the Feature Matrix.

2.1.1. Features

CSG would have to provide all the features mentioned in the Feature Matrix to completely replace TPC without affecting any production processes.

2.1.2. Integration

CSG would have to integrate with all the systems that TPC currently interacts with. On a high level, CSG would have to integrate with:

- CMS (Go Pub)
- SOLR (Search Engine)
- Keychest
- DSAA Clients
- Disney DSAA API Layers

2.1.3. Risks/ Gaps

- **Lack of some of the vital video management features is a big gap** that would result if Disney were to go with CSG as the full turnkey solution. From the Feature Matrix shown above it is clear that CSG cannot support the top ten VMS feature requirements. This would require CSG to build out these features and the timeline and LOE for the completion would then have to be negotiated with them. The possibility of filling in the missing features and finishing successful integration with all the other systems before the DSAA release is highly unlikely.
- There is an additional **risk of data corruption** because of the need to move content currently hosted in the CMS in-house to the CSG database since CSG would need all the content
- With CSG providing a full end-to-end VMS solution, we would be **heavily dependent on vendor support**
- Additional risk of **scope creep** and **quality issues** is introduced with CSG taking on the full development of a new custom VMS since none of the features would be hosted in-house resulting in low visibility

2.2.SHARED SERVICES SOLUTION

This solution proposes that the video management feature be developed in-house at Disney and the remaining playback/ usage services be provided by CSG. The sharing of these services enables Disney to host both content and videos internally with the possibility of combining the CMS and the VMS. Also, since usage/ playback services are not current in-house features, and readily available features in CSG, the transition will be easier. Since this option still considers removing TPC entirely at once, this solution will ideally have to be implemented before DSAA Go-Live.

2.2.1. Features

Referring to the Feature matrix, all the video management features will have to be developed in-house. CSG will have to come up with a robust publishing workflow and a concurrency policy that meets our needs (It has been communicated that CSG has started working on building out features to satisfy our concurrency needs and will be ready by August).

2.2.2. Integration

In this scenario, CSG will have to integrate with the CMS and Keychest through the Client API layers since the third component, which is the VMS, will be built in-house.

2.2.3. Risks/ Gaps

- The aggressive timeline may leave very little room for unforeseen issues and may impact the DSAA timeline
- With a shared solution, Disney will need to develop the gaps that CSG does not provide. From this, integration issues could arise and increase the amount of back-and-forth co-dependent work
- Although CSG would provide the **usage policy side** of the VMS, they still have some features to build out to support DSAA's business needs such as **concurrency**

2.2.4. Level of Effort

This option would require the minimum level of effort of all the options since there will be parallel streams of work going on – one at CSG and one internal at Disney, and hence the timeline would be considerably shortened.

2.3.PHASED CHANGE APPROACH

This approach is an offshoot of the shared services solution in terms of the timeline. This approach allows for an extended timeline while keeping some of the TPC services while gradually moving features in-house or to a different vendor.

The proposed solution extends the timeline so that the teams can focus on one change at a time and plan for any future impacts.

- **Phase 1:** VMS is moved in-house and TPC continues to provide playback/ usage services
- **Phase 2:** Playback/ usage services are moved out of TPC and to another vendor (CSG)

Depending on the results of further analysis, the business and technical teams can decide if DSAA Release can happen between the two phases or before the two phases. This of course introduces risks of changes

post go-live but the increased timeline will allow for the teams to think through the changes and also take into consideration any discoveries post go-live that will help improve a process.

2.3.1. Features & Integration

The Features and Integration points of this approach would be the same as the Shared Services solution.

2.3.2. Risks/ Gaps

- A phased approach would potentially introduce risks with the staggered integration of different systems. By focusing on only one piece of the integration, we wouldn't have a comprehensive view on how future integration points impact it.

3. CONCLUSION

3.1.RECOMMENDATION

The recommended solution is ideally the one with the lowest LOE and has the least amount of risks. The **Shared Services Solution is the option of choice** that falls into this scenario. The effort is split into concurrent development workstreams with Disney creating a video management portion of the VMS custom tailored to DSAA's needs, Medusa 2.0 supporting the video transcoding and packaging, and CSG fine tuning their already available usage management portion to suit our needs. The divide and conquer approach allows each group to focus on their strengths and get to a full solution faster and with smaller shares of potential risks.

The less desirable option with CSG providing us with a full turnkey solution would introduce additional challenges in having them build out the parts of the CMS that they currently don't support. In the end, having a vendor own and control all aspects of the VMS would also put us at the mercy of their ability or inability to respond to change requests quickly and accurately. We would be too dependent on them to support our business needs moving forward.

3.2.TIMELINE

The estimated timeline with proposed milestones is based on the shared services solution.

Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 20	Week 24	Week 28
Project Kickoff	Meet w/ Stakeholders	Re-engage CSG	Finalize PRD	LOE finalization	Development starts	Integrate w/ Medusa 2.0	Development ends	Beta ends
Requirements gathering	Create high level features list	Get PRD/Proposal from CSG	Finalize L1 Architecture	Sprint Planning			Beta starts	Go Live
CSG Demo	Create high level PRD Weekly core team meeting		Contract discussion					

3.3.LEVEL OF EFFORT

Finalizing the Level of Effort depends on various other factors like deciding on a solution, finalizing the architecture, and requirements for the solution and also the vendor capabilities. As reflected in the timeline shown above, once all the information is gathered, the level of effort is planned to be finalized in Week 5.

3.4.DISCUSSION

There are decisions yet to be made on whether to combine CMS and VMS into one system. The team discussed the possibility of opting to go with Go Pub which is the current CMS and that this would be ideal since all the content and video would be in one system instead of separate ones. This will increase the ease of management. However, it is yet to be decided if Go Pub would be the right solution or not.

2.1.CONCEPTUAL ARCHITECTURE DIAGRAM

Given below is a high level architecture diagram for the Shared Services Solution (and also the final state diagram for the Phased Change Approach). As the legend states, the red color denotes the parts that will be supported by open source software/ tools, extended through customization; the blue components would be owned by Medusa2.0. However, depending on the timeline of Medusa, these components along with the ones in Purple will have to be provided by CSG.

